Spatio-Spectral Structure Tensor Total Variation for Hyperspectral Image Denoising and Destriping

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***Abstract*—We propose a novel regularization function, Graph-Aided Spatio-Spectral Total Variation (GASSTV), for hyperspectral (HS) image denoising. Spatio-Spectral Total Variation (SSTV), defined using second-order spatio-spectral differences, is a widely used regularization approach that effectively captures the spatio-spectral piecewise smoothness specific to HS images, enabling robust noise removal. However, SSTV uniformly computes adjacent differences in spatial and spectral dimensions, leading to the corruption of complex spatial structures and abrupt spectral variations in HS images. To address these limitations, we incorporate spatial and spectral graphs that explicitly reflect the spatial and spectral structures into the regularization model. Furthermore, we design a framework to construct these graphs from the noisy HS image. In addition, we formulate the mixed noise removal problem as a convex optimization problem involving GASSTV and develop an efficient algorithm based on the preconditioned primal-dual splitting method to solve it. Experimental results on mixed noise removal demonstrate the superior performance of GASSTV compared with existing HS image regularization models.**

***Index Terms*—Hyperspectral image, denoising, destriping, spatio-spectral regularization, total variation, structure tensor**

I. Introduction

HYPERSPECTRAL (HS) images are three-dimensional data cubes, where each pixel contains a detailed spectrum spanning hundreds of contiguous wavelength bands from the ultraviolet to the near-infrared range. This rich spectral information can visualize materials and phenomena that are invisible in conventional RGB imagery. Furthermore, advanced analytical techniques, such as anomaly detection [1], [2], classification [3]–[5], and unmixing [6], [7], enable the extraction of advanced knowledge from HS images, which can be applied in various fields including agriculture, mineralogy, astronomy, and biotechnology [8]–[11]. Nevertheless, HS images are unavoidably degraded by various types of noise during acquisition arising from sensor defects, photon effects, atmospheric absorption, and other environmental factors [12]–[14]. Such degradations can severely hinder the performance of subsequent analyses, making effective HS image denoising an indispensable preprocessing step for ensuring reliable outcomes.

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Over the past decade, HS image denoising methods have evolved along two main lines: deep neural network (DNN)-based approaches and model-based regularization approaches. DNN-based methods, including convolutional neural networks (CNNs), recurrent models, and transformer-based architectures, have achieved remarkable progress by learning complex spatial–spectral dependencies directly from data. Despite these capabilities, DNNs generally lack strong theoretical underpinnings, making their outputs difficult to interpret and their behavior hard to explain—an important drawback in applications where transparency and reliability are required. In contrast, model-based approaches explicitly encode the physical and statistical properties of HS images into mathematically well-defined formulations. While such methods require careful modeling of the complex nature of HS images, they can provide interpretable and theoretically grounded results. Motivated by these characteristics, this work focuses on advancing the model-based paradigm for HS image denoising.

Model-based approaches for HS image denoising can be broadly categorized into low-rank (LR) and total variation (TV) regularization frameworks. LR-based methods exploit the strong spectral correlations across all bands, modeling HS images as lying in a low-dimensional subspace. This enables the capture of nonlocal spectral relationships and effective suppression of random noise; however, LR models often incur high computational cost and may fail to preserve fine spatial details, especially when strong spatial structures are present. In contrast, TV-based methods are computationally efficient and have shown strong performance by modeling the spatial and spectral piecewise smoothness of HS images. Nevertheless, conventional TV formulations evaluate differences uniformly across adjacent pixels or bands, which limits their ability to preserve complex spatial structures and to capture abrupt spectral variations (spectral jumps), often resulting in the loss of important details. These limitations raise a natural research question: Can we design a TV-based regularization method that retains the efficiency and edge-preserving capabilities of TV while effectively capturing complex spatial structures and spectral jumps for robust HS image denoising and destriping? To answer this question, we would first like to provide a comprehensive review of existing TV-type regularization methods for HS images.

A. Existing TV-type Regularization Methods

The initial TV-based method for HS image denoising is the Spectral-Spatial Adaptive Hyperspectral TV (SSAHTV) [15], were proposed to model spatial piecewise smoothness. Subsequently, Anisotropic Spectral-Spatial TV (ASSTV) 2 was introduced to incorporate spectral differences, capturing the correlation between bands. However, since these methods directly minimize the l_1-norm of first-order differences, they tend to cause over-smoothing, as they cannot sufficiently distinguish between noise and inherent image gradients.

To overcome the over-smoothing issue associated with first-order differences, Spatio-Spectral Total Variation (SSTV) 4 was proposed. SSTV utilizes second-order spatio-spectral differences (i.e., spatial differences of spectral differences), which effectively models the spatial similarity between adjacent bands while avoiding the excessive smoothing typical of first-order methods5555. Due to its effectiveness, SSTV has established itself as a strong baseline and has been extended into various forms.

Several variants have been developed to enhance SSTV's performance. Hybrid Spatio-Spectral TV (HSSTV) 7incorporates first-order spatial differences into the SSTV framework to suppress the generation of continuous artifacts in the spectral direction8. To achieve sharper edges, l_0-l_1 Hybrid TV (l_0-l_1HTV) 9introduces an l_0-norm constraint on spatial differences, allowing for direct control over the number of non-zero gradients (i.e., edge pixels)10. Furthermore, l_0-l_{1-2} SSTV 11 employs the l_{1-2} metric (l_1 minus l_2) instead of the standard l_1 norm. This approach provides a tighter approximation to the l_0 norm, thereby sharpening sparsity and avoiding staircase artifacts 12. Additionally, Spatio-Spectral Structure Tensor TV (S3TTV) captures semi-local spatial structures by evaluating the correlation of difference vectors within semi-local areas using the nuclear norm. However, a common limitation of SSTV and these variants is that they evaluate adjacent differences uniformly in both spatial and spectral directions. Consequently, they often fail to preserve complex spatial structures and spectral jumps (abrupt spectral changes), smoothing them out indistinguishably from noise.

On the other hand, Graph Spatio-Spectral TV (GSSTV) addresses the spatial limitation by weighting the spatial difference operator based on a spatial graph, successfully maintaining complex spatial structures. Nevertheless, GSSTV remains uniform in the spectral direction, failing to capture spectral discontinuities. Furthermore, since the graph weighting is directly incorporated into the SSTV formulation (second-order differences), it suffers from reduced power in determining the solution (regularization ability) compared to methods that explicitly constrain first-order characteristics.

B. Contribution

In this paper, we propose a novel method that incorporates both spatial and spectral graph-based TVs into SSTV, called Graph-Aided Spatio-Spectral Total Variation (GASSTV). The main contributions of this article are listed below:

- We propose a novel regularization method, Graph-Aided Spatio-Spectral Total Variation (GASSTV), which incorporates spatial and spectral graph-based weighted TVs separately from SSTV. These graph-based TVs enhance the noise removal capabilities of SSTV while preserving complex spatial structures and spectral jumps. Furthermore, we robustly construct the spectral graph from the noisy HS image (Refer to GSSTV for constructing the spatial graph).
- We formulate the HS image denoising problem as a constrained convex optimization problem involving GASSTV. To simplify parameter tuning, we impose data fidelity and sparse noise characterization terms as constraints rather than adding them to the objective function.
- We develop an efficient algorithm based on Preconditioned Primal-Dual Splitting Method (P-PDS) [16], [17] to solve our optimization problem. Unlike standard PDS [18], [19], P-PDS automatically determines appropriate stepsizes.

Finally, we demonstrate the effectiveness of GASSTV by comparing it with state-of-the-art HS image regularization methods through HS image denoising experiments.

II. Preliminaries

A. Notations

Throughout this paper, we denote vectors and matrices by boldface lowercase letters (e.g., $\mathbf{x}$) and boldface capital letters (e.g., $\mathbf{X}$), respectively. We treat an HS image, denoted by $\mathbf{u}$ with N_1 vertical pixels, N_2 horizontal pixels, and N_3 bands. We denote the total number of elements in the HS image by $N = N_1N_2N_3$. For a matrix data $\mathbf{X} \in \mathbb{R}^{N_1\times N_2}$, the value at the location (i, j) is denoted by $[\mathbf{X}]_{i,j}$. The ℓ_1-norm and the ℓ_2-norm of a vector $\mathbf{x} \in \mathbb{R}^N$ are defined as $\|\mathbf{x}\|_1 := \sum_{n=1}^N |x_n|$ and $\|\mathbf{x}\|_2 := \sqrt{\sum_{n=1}^N x_n^2}$, respectively, where x_n represents the n-th entry of $\mathbf{x}$. The nuclear norm of a marix, which is the sum of all the singular values, is denoted by $\|\cdot\|_*$. For an HS image $\mathbf{u} \in \mathbb{R}^N$, let $\mathbf{D}_{\mathrm{v}} \in \mathbb{R}^{N\times N}$, $\mathbf{D}_{\mathrm{h}} \in \mathbb{R}^{N\times N}$, and $\mathbf{D}_\lambda \in \mathbb{R}^{N\times N}$ be the forward difference operators along the horizontal, vertical, and spectral directions, respectively, with the periodic boundary condition. Here, a spatial difference operator is denoted by $\mathbf{D}_{\mathrm{vh}} := \begin{pmatrix}\mathbf{D}_{\mathrm{v}}^\top & \mathbf{D}_{\mathrm{h}}^\top\end{pmatrix}^\top \in \mathbb{R}^{2N\times N}$. Using $\mathbf{D}_{\mathrm{v}}$, $\mathbf{D}_{\mathrm{h}}$, and $\mathbf{D}_\lambda$, we denote the second-order spatio-spectral differences by $\mathbf{D}_{\mathrm{v}}\mathbf{D}_\lambda\mathbf{u} \in \mathbb{R}^N$ and $\mathbf{D}_{\mathrm{h}}\mathbf{D}_\lambda\mathbf{u} \in \mathbb{R}^N$. Other notations will be introduced as needed.

B. Proximal Tools

In this chapter, we introduce basic proximal tools that play a central role in the optimization part of our method. Let f be a *proper lower semi-continuous convex function*.[1] Then, for $\gamma > 0$, the *proximity operator* of f is defined by

$$\mathrm{prox}_{\gamma,f}(\mathbf{x}) := \arg\min_{\mathbf{y}\in\mathbb{R}^N} f(\mathbf{y}) + \frac{1}{2\gamma}\|\mathbf{x}-\mathbf{y}\|_2^2. \tag{1}$$

The *Fenchel–Rockafellar conjugate function* f^* of the function f is defined by

$$f^*(\mathbf{x}) := \sup_{\mathbf{y}} \langle \mathbf{x}, \mathbf{y}\rangle - f(\mathbf{y}), \tag{2}$$

where $\langle\cdot,\cdot\rangle$ is the Euclidean inner product. Thanks to a generalization of Moreau's identity [20], the proximity operator of f^* is calculated as

$$\mathrm{prox}_{\gamma,f^*}(\mathbf{x}) = \mathbf{x} - \gamma\,\mathrm{prox}_{\frac{1}{\gamma}f}\left(\frac{1}{\gamma}\mathbf{x}\right). \tag{3}$$

[1] A function $f : \mathbb{R}^N \to (-\infty, \infty]$ is called a proper lower semi-continuous convex function if $\{\mathbf{x} \in \mathbb{R}^N | f(\mathbf{x}) < \infty\}$ is nonempty, $\{\mathbf{x} \in \mathbb{R}^N | f(\mathbf{x}) \leq \alpha\}$ is closed for every $\alpha \in \mathbb{R}$, and $f(\lambda\mathbf{x}+(1-\lambda)\mathbf{y}) \leq \lambda f(\mathbf{x})+(1-\lambda)f(\mathbf{y})$ for every $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$ and $\lambda \in (0, 1)$.

The indicator function of a set $C \subset \mathbb{R}^N$, denoted by ι_C, is defined as

$$\iota_C(\mathbf{x}) := \begin{cases} 0, & \text{if } \mathbf{x} \in C, \\ \infty, & \text{otherwise.} \end{cases} \tag{4}$$

The function ι_C is proper lower semi-continuous convex when C is nonempty and closed convex. The proximity operator of ι_C is equivalent to the projection onto C, as given by

$$\mathrm{prox}_{\iota_C}(\mathbf{x}) = P_C(\mathbf{x}) := \underset{\mathbf{y}\in C}{\mathrm{argmin}} \|\mathbf{y} - \mathbf{x}\|_2. \tag{5}$$

C. Preconditoned Primal-Dual Splitting Method (P-PDS)

The standard PDS [18], [19] and P-PDS [16], on which our algorithm is based, are efficient algorithms for solving the following generic form of convex optimization problems:

$$\begin{aligned} \min_{\substack{\mathbf{x}_1,\ldots,\mathbf{x}_N,\\ \mathbf{y}_1,\ldots,\mathbf{y}_M}} & \sum_{i=1}^{N} f_i(\mathbf{x}_i) + \sum_{j=1}^{M} g_j(\mathbf{y}_j) \\ \text{s.t.} & \begin{cases} \mathbf{y}_1 = \sum_{i=1}^N \mathbf{A}_{1,i}\mathbf{x}_i, \\ \vdots \\ \mathbf{y}_M = \sum_{i=1}^N \mathbf{A}_{M,i}\mathbf{x}_i, \end{cases} \end{aligned} \tag{6}$$

where $f_i (i = 1,\ldots,N)$ and $g_j (j = 1,\ldots,M)$ are lower semi-continuous proper convex functions, $\mathbf{x}_i \in \mathbb{R}^{n_i}$ $(i = 1,\ldots,N)$ are primal variables, $\mathbf{y}_j \in \mathbb{R}^{m_j}$ $(j = 1,\ldots,M)$ are dual variables, and $\mathbf{A}_{j,i} \in \mathbb{R}^{m_j \times n_i}$ $(i = 1,\ldots,N,\ j = 1,\ldots,M)$ are linear operators.

These methods solve Prob. (6) by the following iterative procedures:

$$\left\lfloor \begin{array}{l} \mathbf{x}_1^{(t+1)} \leftarrow \mathrm{prox}_{\gamma_{1,1},f_1}\left(\mathbf{x}_1^{(t)} - \gamma_{1,1}\left(\sum_{j=1}^M \mathbf{A}_{j,1}^\top \mathbf{y}_j^{(t)}\right)\right), \\ \vdots \\ \mathbf{x}_N^{(t+1)} \leftarrow \mathrm{prox}_{\gamma_{1,N},f_N}\left(\mathbf{x}_N^{(t)} - \gamma_{1,N}\left(\sum_{j=1}^M \mathbf{A}_{j,N}^\top \mathbf{y}_j^{(t)}\right)\right), \\ \mathbf{x}_i' = 2\mathbf{x}_i^{(t+1)} - \mathbf{x}_i^{(t)} \ (\forall i = 1,\ldots,N), \\ \mathbf{y}_1^{(t+1)} \leftarrow \mathrm{prox}_{\gamma_{2,1},g_1^*}\left(\mathbf{y}_1^{(t)} - \gamma_{2,1}\left(\sum_{i=1}^N \mathbf{A}_{1,i}\mathbf{x}_i'\right)\right), \\ \vdots \\ \mathbf{y}_M^{(t+1)} \leftarrow \mathrm{prox}_{\gamma_{2,M},g_M^*}\left(\mathbf{y}_M^{(t)} - \gamma_{2,M}\left(\sum_{i=1}^N \mathbf{A}_{M,i}\mathbf{x}_i'\right)\right), \end{array} \right. \tag{7}$$

where $\gamma_{1,i} (i = 1,\ldots,N)$ and $\gamma_{2,j} (j = 1,\ldots,M)$ are the stepsize parameters.

Here, we introduce the convergence property of P-PDS. For the convergence analysis, we define the diagonal matrices of the stepsize parameters as follows:

$$\begin{aligned} \mathbf{\Gamma}_1 &= \mathrm{diag}(\gamma_{1,1}\mathbf{I}_{n_1},\ldots,\gamma_{1,N}\mathbf{I}_{n_N}), \\ \mathbf{\Gamma}_2 &= \mathrm{diag}(\gamma_{2,1}\mathbf{I}_{m_1},\ldots,\gamma_{2,M}\mathbf{I}_{m_M}), \end{aligned} \tag{8}$$

where $\mathbf{I}_{n_i} \in \mathbb{R}^{n_i \times n_i}$ and $\mathbf{I}_{m_j} \in \mathbb{R}^{m_j \times m_j}$ are the identity matrices. We define the linear operator including $\mathbf{A}_{j,i}$ as

$$\mathbf{A} := \begin{bmatrix} \mathbf{A}_{1,1} & \cdots & \mathbf{A}_{1,N} \\ \vdots & \ddots & \vdots \\ \mathbf{A}_{M,1} & \cdots & \mathbf{A}_{M,N} \end{bmatrix}. \tag{9}$$

Then, we state the convergence property of P-PDS.

Theorem 1. *[64, Theorem 1] Let $\mathbf{\Gamma}_1$ and $\mathbf{\Gamma}_2$ be symmetric and positive definite matrices satisfying*

$$\|\mathbf{\Gamma}_1^{\frac{1}{2}} \circ \mathbf{A} \circ \mathbf{\Gamma}_2^{\frac{1}{2}}\|_{\mathrm{op}}^2 < 1. \tag{10}$$

Then, the sequence $\{\mathbf{x}_1^{(t)},\ldots,\mathbf{x}_N^{(t)},\mathbf{y}_1^{(t)},\ldots,\mathbf{y}_M^{(t)}\}$ generated by the procedure in (7) *converges to an optimal solution of Prob.* (6).

The standard PDS needs to adjust the appropriate stepsize parameters to satisfy the convergence conditions (10). On the other hand, P-PDS can automatically determine the stepsize parameters that guarantee convergence [16], [17]. According to [16], we summarize the stepsize design and their convergence property.

Lemma 1. *[64, lemma 2] Let the diagonal matrices $\mathbf{\Gamma}_1, \mathbf{\Gamma}_2$ and the block matrix $\mathbf{A}$ be set as Eq.* (8) *and* (9)*, respectively. In paticular,*

$$\begin{aligned} \gamma_{1,i} &= \frac{1}{\sum_{j=1}^{M}\sum_{l=1}^{m_j} |[\mathbf{A}_{j,i}]_{l,1}|},\ (\forall i = 1,\ldots,N), \\ \gamma_{2,j} &= \frac{1}{\sum_{i=1}^{N}\sum_{k=1}^{n_i} |[\mathbf{A}_{j,i}]_{1,k}|},\ (\forall j = 1,\ldots,M), \end{aligned} \tag{11}$$

then the inquality in (10) *holds.*

D. Weighted Spatial Difference Operator based on Spatial Graph by GSSTV [21]

Since we only obtain a noisy HS image in real situations, we construct a spatial graph via a guide image computed by

$$\mathbf{v}'(i,j) := \frac{1}{N_3}\sum_{c=1}^{N_3} \mathbf{v}(i,j,c), \tag{12}$$

where $\mathbf{v}$ is a given HS image. The guide image is averaged in the spectral direction to reduce noise while maintaining the spatial structure. Next, we consider to construct a weighted graph $\mathcal{G}_{\mathrm{sp}}(\mathbf{v}', \mathcal{E}_{\mathrm{sp}}, \mathbf{W}_{\mathrm{sp}}')$ with the edges $e_{p,q} \in \mathcal{E}_{\mathrm{sp}}$ (p and q are indices of pixels $(1 \leq p < q \leq N_1N_2)$). The weight matrix $\mathbf{W}_{\mathrm{sp}}' \in \mathbb{R}^{|\mathcal{E}_{\mathrm{sp}}|\times|\mathcal{E}_{\mathrm{sp}}|}$ is a diagonal matrix whose entries are the weights $w_{e_{p,q}} \in (0,1]$ assigned to the edges $e_{p,q}$, defined as

$$w_{e_{p,q}} := \exp\left(-\frac{({v'}_p - {v'}_q)^2}{2\sigma_{\mathrm{sp}}^2}\right), \tag{13}$$

where σ_{sp} is a parameter. The value of $w_{e_{p,q}}$ is large as the correlation between ${v'}_p$ and ${v'}_q$ is higher. We also introduce the incidence matrix $\mathbf{D} \in \mathbb{R}^{|\mathcal{E}_{\mathrm{sp}}|\times N_1N_2}$ for a graph $\mathcal{G}_{\mathrm{sp}}$ as follows:

$$\mathbf{D}_{e_{p,q},k} = \begin{cases} -1 & \text{if } p = k, \\ 1 & \text{if } q = k, \quad (\forall k = 1,\ldots,N_1N_2). \\ 0 & \text{otherwise,} \end{cases} \tag{14}$$

To apply the spatial graph total variation to the entire HS image, we extend the weight matrix $\mathbf{W}_{\mathrm{sp}}'$ and the incidence matrix $\mathbf{D}$ to block diagonal matrices $\mathbf{W}_{\mathcal{G}_{\mathrm{sp}}} \in \mathbb{R}^{N_3|\mathcal{E}_{\mathrm{sp}}|\times N_3|\mathcal{E}_{\mathrm{sp}}|}$ and $\mathbf{D}_{\mathcal{G}_{\mathrm{sp}}} \in \mathbb{R}^{N_3|\mathcal{E}_{\mathrm{sp}}|\times N}$, each consisting of N_3 repetitions of $\mathbf{W}_{\mathrm{sp}}'$ and $\mathbf{D}$, respectively, defined as:

$$\mathbf{W}_{\mathcal{G}_{\mathrm{sp}}} := \mathrm{diag}(\mathbf{W}_{\mathrm{sp}}',\ldots,\mathbf{W}_{\mathrm{sp}}'), \quad \mathbf{D}_{\mathcal{G}_{\mathrm{sp}}} := \mathrm{diag}(\mathbf{D},\ldots,\mathbf{D}). \tag{15}$$

Then, the weighted spatial graph difference operator for an HS image is given by $\mathbf{W}_{\mathcal{G}_{\text{sp}}}\mathbf{D}_{\mathcal{G}_{\text{sp}}}$.

III. Proposed Method

We propose a novel regularization function incorporating spatial and spectral graphs. For the spatial graph, we utilize the construction described in Sec. II-D. In the following, we first describe define the spectral graph and then present the proposed regularization function. Subsequently, we formulate an HS image denoising problem as a constrained convex optimization problem involving the GASSTV regularization function and finally develpop an algorithm to solve it.

A. Spectral Graph-based Weighted Difference Operator

In constructing the spectral graph from a noisy HS image, we assume that when the HS image is segmented, each segment contains similar spectral characteristics. Based on this assumption, we first segment the noisy HS image into K regions and then construct a spectral graph for each segment using its representative spectrum. To obtain the segments, we compute the guide image $\mathbf{v}'$ that reflects the spatial structure of the target HS image according to Eq. (12). Using the k-means clustering algorithm [22] on the guide image $\mathbf{v}'$, we obtain K segments $\{\mathcal{S}_k\}_{k=1}^{K}$ by solving the following minimization problem:

$$\min_{\{\mathcal{S}_k\}\{\mu_k\}} \sum_{k=1}^{K} \sum_{(i,j)\in\mathcal{S}_k} \|\mathbf{v}'(i,j) - \mu_k\|_2^2, \tag{16}$$

where μ_k represents the centroid of the k-th segment. For each segment $\mathcal{S}_k$, we compute a representative spectrum $\mathbf{r}_k \in \mathbb{R}^{N_3}$ by averaging the spectral vectors of the noisy HS image within the segment:

$$\mathbf{r}_k := \frac{1}{|\mathcal{S}_k|} \sum_{(i,j)\in\mathcal{S}_k} \mathbf{v}(i,j,:), \tag{17}$$

where $\mathbf{v}(i,j,:)$ denotes the spectral vector with length N_3 at location (i,j). By averaging the spectral vectors within each segment, the spectra of the segment can be effectively characterized while reducing noise.

Following the calculation of the representative spectra, we construct a spectral graph for each segment to explicitly capture spectral variations including smoothness and spectral jumps. We design the graph nodes to represent the spectral adjacent differences and connect the nodes exhibiting similar variations. First, for the k-th segment, we calculate the spectral difference vector $\mathbf{d}_k \in \mathbb{R}^{N_3}$, where the p-th element $\mathbf{d}_{k,p}$ is defined as:

$$\mathbf{d}_{k,p} := \mathbf{r}_{k,p+1} - \mathbf{r}_{k,p}, \quad (1 \le p \le N_3 - 1) \tag{18}$$

where $\mathbf{r}_{k,p}$ is the p-th element of the representative spectrum $\mathbf{r}_k$. Here, we assume the Neumann boundary condition (i.e., $\mathbf{r}_{k,N_3} = 0$). Next, we define a graph $\mathcal{G}_\lambda^{(k)}$ where the nodes correspond to the difference indices. To construct the edges, we employ the M-nearest neighbor (M-NN) strategy based on the distance between the gradient values. Let $\mathcal{N}_k$ denote the set of the M nearest neighbors of the node p determined by the Euclidean distance $|\mathbf{d}_{k,p} - \mathbf{d}_{k,q}|$. The weight $w_{p,q}^{(k)}$ is defined as: for $1 \le p < q \le N_3 - 1$,

$$w_{p,q}^{(k)} := \begin{cases} \exp\left(-\frac{(\mathbf{d}_{k,p} - \mathbf{d}_{k,q})^2}{2\sigma_\lambda^2}\right) & \text{if } q \in \mathcal{N}_k \text{ or } p \in \mathcal{N}_k, \\ 0 & \text{otherwise,} \end{cases} \tag{19}$$

where σ_λ is a scaling parameter.

The Laplacian matrix $\mathbf{L}^{(k)} \in \mathbb{R}^{N_3 \times N_3}$ is then computed as $\mathbf{L}^{(k)} := \mathbf{D}^{(k)} - \mathbf{W}_\lambda k$. Finally, we extend these local Laplacians to the global spectral graph Laplacian $\mathbf{L}_{\mathcal{G}_\lambda} \in \mathbb{R}^{N \times N}$. Let $l_p \in 1, \ldots, K$ denote the segment label of the p-th spatial pixel. The global matrix is constructed as a block diagonal matrix:

$$\mathbf{L}_{\mathcal{G}_\lambda} := \text{diag}(\mathbf{L}^{(l_1)}, \mathbf{L}^{(l_2)}, \ldots, \mathbf{L}^{(l_{N_1 N_2})}). \tag{20}$$

B. GASSTV

Incorporating both the spatial graph $\mathcal{G}_{\text{sp}}$ and the spectral graph $\mathcal{G}_\lambda$, our *Graph-Aided Spatio-Spectral Total Variation* (GASSTV) is defined as follows:

$$\begin{aligned} \text{GASSTV}(\mathbf{u}) &:= \|\mathbf{D}_{\text{vh}}\mathbf{D}_\lambda\mathbf{u}\|_1 + \frac{\omega}{2}(\mathbf{D}_\lambda\mathbf{u})^\top \mathbf{L}_{\mathcal{G}_\lambda}\mathbf{D}_\lambda\mathbf{u}, \\ &\text{s.t. } \|\mathbf{W}_{\mathcal{G}_{\text{sp}}}\mathbf{D}_{\mathcal{G}_{\text{sp}}}\mathbf{u}\|_1 \le \rho_{\text{sp}}, \end{aligned} \tag{21}$$

where $\mathbf{D}_{\text{vh}} := \begin{pmatrix}\mathbf{D}_{\text{v}}^\top & \mathbf{D}_{\text{h}}^\top\end{pmatrix}^\top \in \mathbb{R}^{2N \times N}$ is the spatial difference operator, ω and ρ_{sp} are the parameters. The formulation consists of three key components:

1) Spatio-Spectral Total Variation (SSTV): The first term in (21) is the standard SSTV [21], which imposes sparsity on the second-order spatio-spectral differences. This term promotes piecewise smoothness simultaneously across spatial and spectral dimensions, serving as the baseline for noise suppression.
2) Spectral Graph Laplacian Regularizer: The second term in (21) introduces the spectral graph regularization. We employ the Graph Laplacian quadratic form based on the constructed spectral graph $\mathcal{G}_\lambda$. This term imposes smoothness on the spectral variations over the graph structure. By minimizing the weighted differences of gradients between connected nodes, it ensures that spectral bands exhibiting similar fluctuation trends in the representative spectra maintain consistent variations in the restored image. Consequently, this mechanism effectively preserves smooth spectral curves while explicitly maintaining inherent spectral jumps, which would otherwise be smoothed out by uniform regularization.
3) Spatial Graph Total Variation Constraint: The inequality constraint (22) incorporates the spatial structural information. Here, we use the weighted Graph Total Variation based on the l_1-norm of the first-order spatial differences. The use of the l_1-norm is crucial as it promotes sparsity in the gradient domain, thereby strictly preserving sharp edges and complex spatial structures without blurring. Furthermore, by formulating this term as a hard constraint rather than a penalty in the objective function, we can directly control the degree of spatial smoothness. The upper bound ρ_{sp} is adaptively

determined by the signal intensity of the observed image $\mathbf{v}$ and the structure of the guide image $\mathbf{v}'$ as:

$$\rho_{\text{sp}} := \omega_{\text{sp}} \frac{\|\mathbf{v}\|_1}{N_1 N_2} \|\mathbf{W}_{\text{sp}}^{'} \mathbf{D}\mathbf{v}'\|_1, \tag{22}$$

where $\omega_{\text{sp}} > 0$ is a parameter. This definition effectively scales the spatial variation of the guide image by the sum of average intensities of all bands in the observed image $\mathbf{v}'$.

Consequently, by integrating these complementary regularization functions, GASSTV effectively suppresses mixed noise while faithfully preserving complex spatial structures and spectral jumps.

C. Problem Formulation

An observed HS image $\mathbf{v} \in \mathbb{R}^N$ contaminated by mixed noise is modeled by

$$\mathbf{v} = \bar{\mathbf{u}} + \bar{\mathbf{s}} + \bar{\mathbf{t}} + \mathbf{n}, \tag{23}$$

where $\bar{\mathbf{u}}$, $\bar{\mathbf{s}}$, $\bar{\mathbf{t}}$, and $\mathbf{n}$ represent a clean HS image, sparse noise such as outliers, stripe noise and random noise, respectively.

Based on the observation model (23), we formulate HS image denoising problem by GASSTV as a constrained convex optimization problem with the following form:

$$\min_{\mathbf{u},\mathbf{s},\mathbf{t}} \text{GASSTV}(\mathbf{u}) \quad \text{s.t.} \begin{cases} \mathbf{s} \in B_{1,\alpha}, \\ \mathbf{t} \in B_{1,\beta}, \\ \mathbf{D}_{\text{v}}\mathbf{t} = \mathbf{0}, \\ \mathbf{u} + \mathbf{s} + \mathbf{t} \in B_{2,\varepsilon}^{\mathbf{v}}, \\ \mathbf{u} \in R_{\underline{\mu},\bar{\mu}}, \end{cases} \tag{24}$$

where

$$B_{1,\alpha} := \{\mathbf{x} \in \mathbb{R}^N \mid \|\mathbf{x}\|_1 \leq \alpha\}, \tag{25}$$

$$B_{1,\beta} := \{\mathbf{x} \in \mathbb{R}^N \mid \|\mathbf{x}\|_1 \leq \beta\}, \tag{26}$$

$$B_{2,\varepsilon}^{\mathbf{v}} := \{\mathbf{x} \in \mathbb{R}^N \mid \|\mathbf{x} - \mathbf{v}\|_2 \leq \varepsilon\}, \tag{27}$$

$$R_{\underline{\mu},\bar{\mu}} := \{\mathbf{x} \in \mathbb{R}^N \mid \underline{\mu} \leq x_i \leq \bar{\mu} \, (i = 1, \ldots, N)\}. \tag{28}$$

The first constraint characterizes sparse noise $\mathbf{s}$ with the zero-centered ℓ_1-ball of the radius $\alpha > 0$. The second constraint controls the intensity of stripe noise $\mathbf{t}$ and the third constraint captures the vertical flatness property by imposing zero to the vertical gradient of $\mathbf{t}$. These constraints effectively characterize stripe noise [23]. The fourth constraint serves as data-fidelity with the $\mathbf{v}$-centered ℓ_2-ball of the radius $\varepsilon > 0$. The fifth constraint is a box constraint with $\underline{\mu} < \bar{\mu}$ which represents the dynamic range of $\mathbf{u}$. For normalized HS images, we can set $\underline{\mu} = 0$ and $\bar{\mu} = 1$.

Using the first, second, and fourth constraints instead of adding terms to the objective function makes it much easier to adjust the parameters α, β, and ε. This is because by expressing multiple terms as constraints, rather than adding them to the objective function, the hyperparameters associated with each term are converted to be independent of each other, and appropriate parameters can be determined without interdependence. Such advantage has been addressed, e.g., in [24]–[28].

Algorithm 1 P-PDS-based solver for (18)

Require: $\mathbf{u}^{(0)}, \mathbf{s}^{(0)}, \mathbf{t}^{(0)}, \mathbf{y}_1^{(0)}, \mathbf{y}_2^{(0)}, \mathbf{y}_3^{(0)}, \mathbf{y}_4^{(0)}$
Ensure: $\mathbf{u}^{(t)}$

while A stopping criterion is not satisfied **do**
  $\tilde{\mathbf{y}} \leftarrow \mathbf{D}_\lambda^\top \mathbf{D}_{\text{vh}}^\top \mathbf{y}_1^{(t)} + \mathbf{D}_{\mathcal{G}_{\text{sp}}}^\top \mathbf{W}_{\mathcal{G}_{\text{sp}}}^\top \mathbf{y}_2^{(t)} + \mathbf{y}_4^{(t)}$
  $\mathbf{u}^{(t+1)} \leftarrow P_{R_{\underline{\mu},\bar{\mu}}} \left(\mathbf{u}^{(t)} - \gamma_{\mathbf{u}}(\omega \mathbf{D}_\lambda^\top \mathbf{L}_{\mathcal{G}_\lambda} \mathbf{D}_\lambda \mathbf{u}^{(t)} + \tilde{\mathbf{y}})\right)$;
  $\mathbf{s}^{(t+1)} \leftarrow P_{B_{1,\alpha}} \left(\mathbf{s}^{(t)} - \gamma_{\mathbf{s}} \mathbf{y}_4^{(t)}\right)$;
  $\mathbf{t}^{(t+1)} \leftarrow P_{B_{1,\beta}} \left(\mathbf{t}^{(t)} - \gamma_{\mathbf{t}} \left(\mathbf{D}_{\text{v}}^\top \mathbf{y}_3^{(t)} + \mathbf{y}_4^{(t)}\right)\right)$;
  $\mathbf{u}^{'} \leftarrow 2\mathbf{u}^{(t+1)} - \mathbf{u}^{(t)}$;
  $\mathbf{s}^{'} \leftarrow 2\mathbf{s}^{(t+1)} - \mathbf{s}^{(t)}$;
  $\mathbf{t}^{'} \leftarrow 2\mathbf{t}^{(t+1)} - \mathbf{t}^{(t)}$;
  $\mathbf{y}_1^{'} \leftarrow \mathbf{y}_1^{(t)} + \gamma_{\mathbf{y}} \mathbf{D}_{\text{vh}} \mathbf{D}_\lambda \mathbf{u}^{'}$;
  $\mathbf{y}_1^{(t+1)} \leftarrow \mathbf{y}_1^{'} - \gamma_{\mathbf{y}} \operatorname{prox}_{\gamma_{\mathbf{y}} \|\cdot\|_1} \left(\frac{1}{\gamma_{\mathbf{y}}} \mathbf{y}_1^{'}\right)$;
  $\mathbf{y}_2^{'} \leftarrow \mathbf{y}_2^{(t)} + \gamma_{\mathbf{y}} \mathbf{W}_{\mathcal{G}_{\text{sp}}} \mathbf{D}_{\mathcal{G}_{\text{sp}}} \mathbf{u}^{'}$;
  $\mathbf{y}_2^{(t+1)} \leftarrow \mathbf{y}_2^{'} - \gamma_{\mathbf{y}} P_{B_{1,\rho_{\text{sp}}}} \left(\frac{1}{\gamma_{\mathbf{y}}} \mathbf{y}_2^{'}\right)$;
  $\mathbf{y}_3^{(t+1)} \leftarrow \mathbf{y}_3^{(t)} + \gamma_{\mathbf{y}} \mathbf{D}_{\text{v}} \mathbf{t}^{'}$;
  $\mathbf{y}_4^{'} \leftarrow \mathbf{y}_4^{(t)} + \gamma_{\mathbf{y}} \left(\mathbf{u}^{'} + \mathbf{s}^{'} + \mathbf{t}^{'}\right)$;
  $\mathbf{y}_4^{(t+1)} \leftarrow \mathbf{y}_4^{'} - \gamma_{\mathbf{y}} P_{B_{2,\varepsilon}^{\mathbf{v}}} \left(\frac{1}{\gamma_{\mathbf{y}}} \mathbf{y}_4^{'}\right)$;
  $t \leftarrow t + 1$;
end while

D. Optimization

We develop an efficient solver for Prob. (24) based on P-PDS [16]. Using the indicator functions, we rewrite Prob. (24) into an equivalent form:

$$\begin{aligned} \min_{\substack{\mathbf{u},\mathbf{s},\mathbf{t}, \\ \mathbf{y}_1,\mathbf{y}_2,\mathbf{y}_3,\mathbf{y}_4}} \ & \|\mathbf{y}_1\|_1 + \frac{\omega}{2}(\mathbf{D}_\lambda \mathbf{u})^\top \mathbf{L}_{\mathcal{G}_\lambda} \mathbf{D}_\lambda \mathbf{u} + \iota_{B_{1,\rho_{\text{sp}}}}(\mathbf{y}_2) \\ & + \iota_{B_{1,\alpha}}(\mathbf{s}) + \iota_{B_{1,\beta}}(\mathbf{t}) + \iota_{\{\mathbf{0}\}}(\mathbf{y}_3) \\ & + \iota_{B_{2,\varepsilon}^{\mathbf{v}}}(\mathbf{y}_4) + \iota_{R_{\underline{\mu},\bar{\mu}}}(\mathbf{u}) \\ \text{s.t.} & \begin{cases} \mathbf{y}_1 = \mathbf{D}_{\text{vh}} \mathbf{D}_\lambda \mathbf{u}, \\ \mathbf{y}_2 = \mathbf{W}_{\mathcal{G}_{\text{sp}}} \mathbf{D}_{\mathcal{G}_{\text{sp}}} \mathbf{u}, \\ \mathbf{y}_3 = \mathbf{D}_{\text{v}} \mathbf{t}, \\ \mathbf{y}_4 = \mathbf{u} + \mathbf{s} + \mathbf{t}. \end{cases} \end{aligned} \tag{29}$$

Let $\mathbf{u}$, $\mathbf{s}$, and $\mathbf{t}$ be the primal variables and $\mathbf{y}_1, \ldots, \mathbf{y}_4$ be the dual variables. Then, by defining,

$$\begin{aligned} \varphi_1(\mathbf{u}) &:= \frac{\omega}{2}(\mathbf{D}_\lambda \mathbf{u})^\top \mathbf{L}_{\mathcal{G}_\lambda} \mathbf{D}_\lambda \mathbf{u}, \\ f_1(\mathbf{u}) &:= \iota_{R_{\underline{\mu},\bar{\mu}}}(\mathbf{u}), \\ f_2(\mathbf{s}) &:= \iota_{B_{1,\alpha}}(\mathbf{s}), \\ f_3(\mathbf{t}) &:= \iota_{B_{1,\beta}}(\mathbf{t}), \\ g_1(\mathbf{y}_1) &:= \|\mathbf{y}_1\|_1, \\ g_2(\mathbf{y}_2) &:= \iota_{B_{1,\rho_{\text{sp}}}}(\mathbf{y}_2), \\ g_3(\mathbf{y}_3) &:= \iota_{\{\mathbf{0}\}}(\mathbf{y}_3), \\ g_4(\mathbf{y}_4) &:= \iota_{B_{2,\varepsilon}^{\mathbf{v}}}(\mathbf{y}_4), \end{aligned} \tag{30}$$

Prob. (**??**) is reduced to Prob. (6). Therefore, P-PDS is applicable to Prob. (**??**). Prob. (29) can be solved by P-PDS [16] given by Alg. 1. The proximity operator of $\|\cdot\|_1$ is calculated by

$$[\mathrm{prox}_{\gamma\|\cdot\|_1}(\mathbf{x})]_i = \mathrm{sgn}(x_i)\max\{0, |x_i| - \gamma\}, \quad (31)$$

which is equivalent to soft thresholding. The projections onto $B_{2,\varepsilon}^{\mathbf{v}}$ and $R_{\underline{\mu},\bar{\mu}}$ are calculated by

$$P_{B_{2,\varepsilon}^{\mathbf{v}}}(\mathbf{x}) = \begin{cases} \mathbf{x}, & \text{if } \mathbf{x} \in B_{2,\varepsilon}^{\mathbf{v}}, \\ \mathbf{v} + \frac{\varepsilon(\mathbf{x}-\mathbf{v})}{\|\mathbf{x}-\mathbf{v}\|_2}, & \text{otherwise}, \end{cases} \quad (32)$$

$$[P_{R_{\underline{\mu},\bar{\mu}}}(\mathbf{x})]_i = \begin{cases} \underline{\mu}, & \text{if } x_i < \underline{\mu}, \\ \bar{\mu}, & \text{if } x_i > \bar{\mu}, \\ x_i, & \text{otherwise}. \end{cases} \quad (33)$$

The projection onto $B_{1,\alpha}$ can be efficiently computed by a fast ℓ_1-ball projection algorithm [29]. According to [17], we set the stepsize parameters as

$$\gamma_{\mathbf{u}} = \frac{1}{49 + 2\omega \max_k \sigma_{\max}(\mathbf{L}^{(k)})}, \\ \gamma_{\mathbf{s}} = 1,\ \gamma_{\mathbf{t}} = \frac{1}{5},\ \gamma_{\mathbf{y}} = \frac{1}{3}, \quad (34)$$

where $\sigma_{\max}(\mathbf{L}^{(k)})$ is the maximum singular value of the Laplacian matrix of the segment k.

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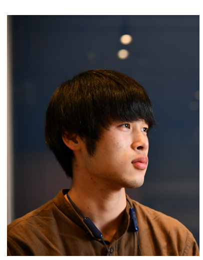

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